

Introduction to Computational Fluid Dynamics

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Outline

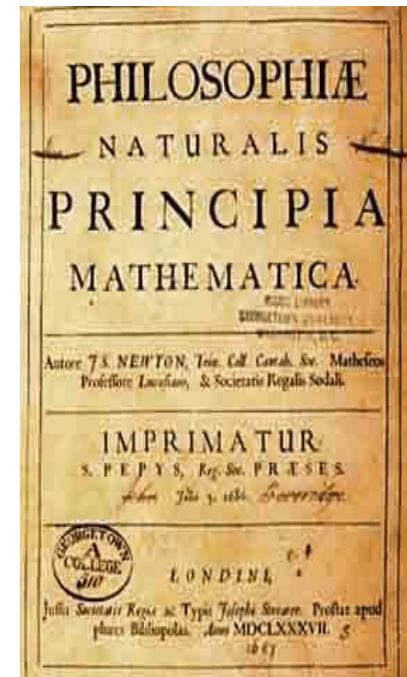
- **Review of Fluid Mechanics**
- **Solution Methods for Incompressible Navier-Stokes Equations**
- **Solution Methods for Compressible Navier-Stokes Equations**

Review of Fluid Mechanics

History of Fluid Mechanics

● History

- Archimedes (287-212 BC)
 - Introduced some basic ideas in **fluid statics**
- Leonardo da Vinci (1452-1519)
 - Observed and drew **sketches of complex flows over objects** in streams
- Isaac Newton (1642-1727)
 - Devoted Book II of his Principia Mathematica (1687) exclusively to the examination of fluid dynamics and fluid statics
 - **Newton's law of viscosity** → Newtonian fluid



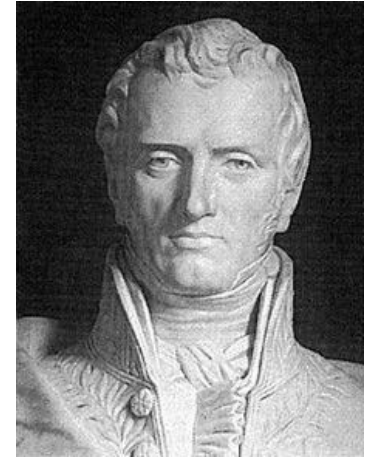
History of Fluid Mechanics

- Daniel Bernoulli (1700-82), Jean le Rond d'Alembert (1717-83), and Leonhard Euler (1707-83)
 - A mathematical formulation of a fluid flow took shape with their contributions
 - Euler described flow in terms of spatially varying three-dimensional pressure and velocity and modeled **the flow as a continuous collection of infinitesimally small fluid elements**
 - By applying the basic principles of mass conservation and Newton's second law, **Euler obtained two coupled, nonlinear PDEs involving the flow fields of pressure and velocity (1757)**
 - **Euler did not account for the effect of friction** acting on the motion of the fluid elements – that is, he ignored viscosity



History of Fluid Mechanics

- Claude-Louis Navier (1822) and George Stokes (1845)
 - The Euler equations were modified to account for **the effect of internal friction** within a flow field
 - The resulting equations, a system of even more elaborate nonlinear PDEs, are now called **the Navier-Stokes equations**
 - **The inability to solve the Navier-Stokes equations** for most practical flow problems was particularly frustrating to those investigators interested in calculating **the friction shear force on a surface immersed in a flow**
 - This difficulty became acute at the beginning of 20th century, with the invention of the first practical airplane by Orville and Wilbur Wright and with the subsequent need to calculate **the lift and drag on airplanes**

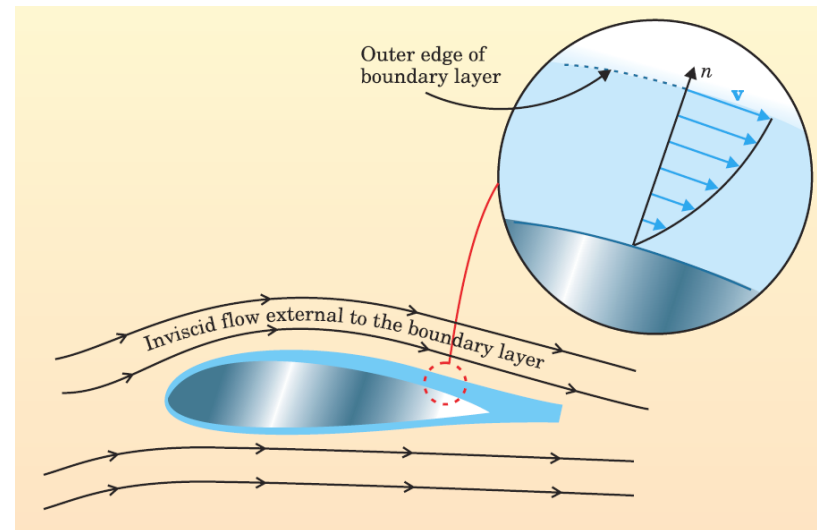
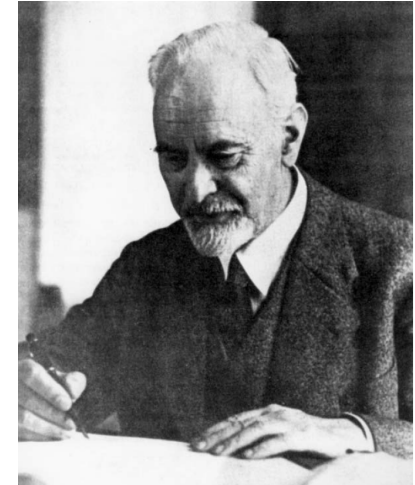
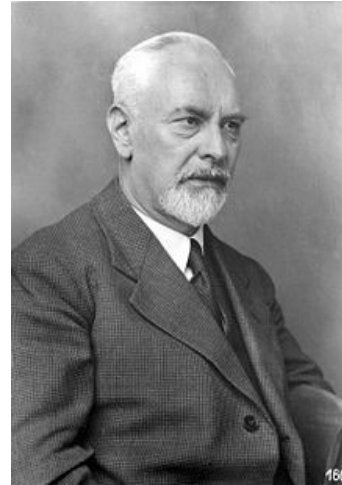


History of Fluid Mechanics



History of Fluid Mechanics

- Ludwig Prandtl (1904)
 - **The boundary layer** concept
 - Seminal presentation at Heidelberg, **“On the motion of fluids with very little friction”** which was one of the most important fluid dynamics paper ever written
 - An effect of friction was to cause the fluid immediately adjacent to the surface to stick to the surface and that **frictional effects were experienced only in a boundary layer**, a thin region near the surface
 - **Outside the boundary layer, the flow was essentially the inviscid flow** that had been studied for the previous two centuries



Available from <http://dx.doi.org/10.1063/1.2169443>

The Boundary-Layer Concept

- **Boundary layer**

- Theoretical hydrodynamics (d'Alembert's paradox, 1752) vs. empirical hydraulics
- **Navier-Stokes** equations (1822, 1845)
- Stokes (1851) solved **the drag on a sphere** in Stokes flow (low speed)

$$F_D = 3\pi\mu VD; \quad C_D = \frac{24}{Re}$$

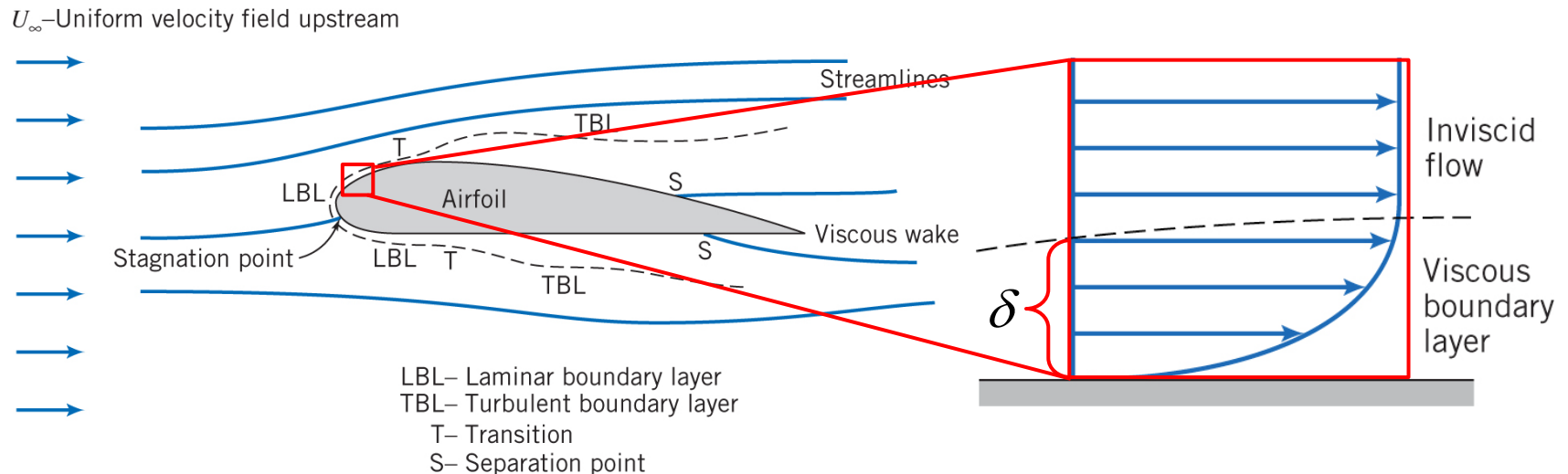
- For increasing Reynolds numbers, the viscous Navier–Stokes equations converge towards the inviscid Euler equations

$$\begin{aligned} \frac{\partial u^*}{\partial x^*} + \frac{\partial v^*}{\partial y^*} &= 0 \\ u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} &= -\frac{\partial p^*}{\partial x^*} + \frac{1}{Re} \left(\frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}} \right) \\ u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} &= -\frac{\partial p^*}{\partial y^*} + \frac{1}{Re} \left(\frac{\partial^2 v^*}{\partial x^{*2}} + \frac{\partial^2 v^*}{\partial y^{*2}} \right) \end{aligned}$$

The Boundary-Layer Concept

- **Boundary layer**

- Ludwig Prandtl (1904) suggested **a thin viscous boundary layer**
- Even though friction is negligible in general for high Re flows, there exists a thin viscous boundary layer
- Friction is significant and velocity variation within the layer is huge (no slip at the wall $\sim$ outflow velocity)



- <http://web.mit.edu/hml/ncfmf.html>, Fundamentals-Boundary Layers

Basic Relations

Reynolds Transport Theorem

For any vector or scalar function

$$\frac{D}{Dt} \int_{V_{sys}} \phi dV = \frac{\partial}{\partial t} \int_{CV} \phi dV + \int_{CS} \phi (\mathbf{u} \cdot \mathbf{n}) dS$$

Rate of change
in system

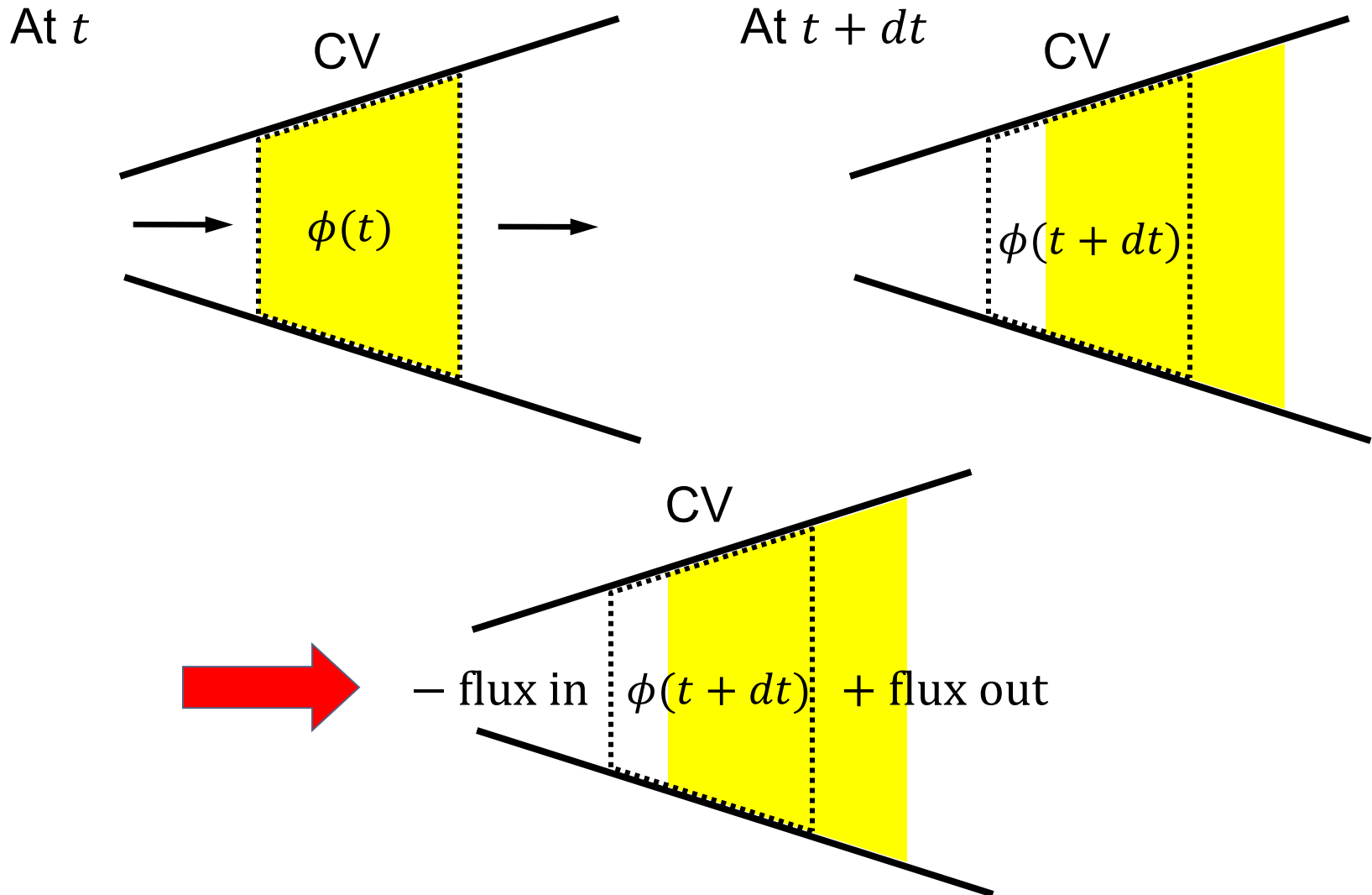
Rate of change
in control volume

Flux through
control surface

Material derivative

$$\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + \mathbf{u} \cdot \nabla\phi = \frac{\partial\phi}{\partial t} + u \frac{\partial\phi}{\partial x} + v \frac{\partial\phi}{\partial y} + w \frac{\partial\phi}{\partial z}$$

Basic Relations

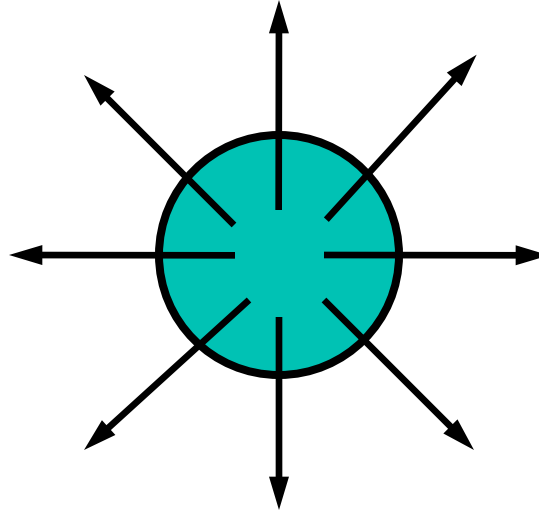


Basic Relations

The Divergence (Gauss) Theorem:

Conversion of surface integral to volume integral

$$\int_{CS} (\boldsymbol{\Phi} \cdot \mathbf{n}) dS = \int_{CV} (\nabla \cdot \boldsymbol{\Phi}) dV$$



Basic Relations

Reynolds Transport Theorem (RTT)

System integral to control volume integral

$$\begin{aligned}\frac{D}{Dt} \int_{V_{sys}} \phi dV &= \frac{\partial}{\partial t} \int_{CV} \phi dV + \int_{CS} \phi (\mathbf{u} \cdot \mathbf{n}) dS \\ &= \int_{CV} \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{u}) \right) dV\end{aligned}$$

Conservation Equations

Conservation of Mass (Continuity equation)

If $\phi(\mathbf{x}, t) = \rho(\mathbf{x}, t)$, Reynolds transport theorem (RTT) yields

$$\frac{D}{Dt} \int_{V_{sys}} \rho dV = 0 = \frac{\partial}{\partial t} \int_{CV} \left(\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right) dV$$

Since the equation holds for arbitrary control volume,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

Conservation Equations

Or in Lagrangian form,

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) &= \frac{\partial \rho}{\partial t} + \mathbf{u} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{u} \\ &= \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0\end{aligned}$$

Continuity Equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad \text{or} \quad \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0$$

Conservation Equations

Conservation of Momentum (Momentum equation)

If $\phi(\mathbf{x}, t) = \rho \mathbf{u}(\mathbf{x}, t)$, RTT yields

$$\begin{aligned} \frac{D}{Dt} \int_{V_{sys}} \rho \mathbf{u} dV &= \int_{CV} \left(\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) \right) dV \\ &= \int_{CS} (\mathbf{T} \cdot \mathbf{n}) dS + \int_{CV} \rho \mathbf{f} dV \\ &= \int_{CV} (\nabla \cdot \mathbf{T} + \rho \mathbf{f}) dV \end{aligned}$$

Divergence theorem

Surface force (stress) Body force (e.g. gravity)

Conservation Equations

Conservation of Momentum (Momentum equation)

$$\frac{\partial}{\partial t}(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = \nabla \cdot \mathbf{T} + \rho \mathbf{f}$$

Alternatively, subtracting $\mathbf{u} \cdot (\text{continuity})$

$$\begin{aligned} \frac{\partial}{\partial t}(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \mathbf{u} \left(\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right) \\ = \rho \frac{\partial \mathbf{u}}{\partial t} + \rho \mathbf{u} \cdot \nabla \mathbf{u} = \rho \frac{D\mathbf{u}}{Dt} = \nabla \cdot \mathbf{T} + \rho \mathbf{f} \end{aligned}$$

$$\rho \frac{D\mathbf{u}}{Dt} = \nabla \cdot \mathbf{T} + \rho \mathbf{f}$$

Conservation Equations

Constitutive Relations – Stress Tensor

$$\mathbf{T} = \left[-p + \left(\kappa - \frac{2}{3}\mu \right) \nabla \cdot \mathbf{u} \right] \mathbf{U} + 2\mu \mathbf{D}$$

Unit tensor

Bulk viscosity = 0 (Stokes assumption)

$$\mathbf{D} = \frac{1}{2} [(\nabla \mathbf{u}) + (\nabla \mathbf{u})^T] : \text{Deformation tensor}$$

In tensor notation (Einstein notation)

$$T_{ij} = -p\delta_{ij} + \left(\kappa - \frac{2}{3}\mu \right) \delta_{ij} \frac{\partial u_k}{\partial x_k} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

Conservation Equations

Conservation of Momentum (Final):

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \left(\left(\kappa - \frac{2}{3}\mu \right) \nabla \cdot \mathbf{u} \right) + \nabla \cdot (2\mu \mathbf{D}) + \rho \mathbf{f}$$

$$\mathbf{D} = \frac{1}{2} [(\nabla \mathbf{u}) + (\nabla \mathbf{u})^T]$$

Conservation Equations

Conservation of Energy (Energy equation)

If $\Phi(\mathbf{x}, t) = \rho(e(\mathbf{x}, t) + \mathbf{u} \cdot \mathbf{u}/2)$, RTT yields

$$\begin{aligned} \frac{D}{Dt} \int_{V_{sys}} \rho \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) dV &= \int_{CV} \left[\frac{\partial}{\partial t} \left\{ \rho \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \right\} + \nabla \cdot \left\{ \rho \mathbf{u} \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \right\} \right] dV \\ &= - \int_{CS} (\mathbf{q} \cdot \mathbf{n}) dS + \int_{CS} \mathbf{u} \cdot (\mathbf{n} \cdot \mathbf{T}) dS + \int_{CV} \rho \mathbf{u} \cdot \mathbf{f} dV \\ &\quad \text{Heat flux} \qquad \text{Work by stress} \qquad \text{Body force work} \\ &\stackrel{\text{Divergence theorem}}{=} \int_{CV} [-\nabla \cdot \mathbf{q} + \nabla \cdot (\mathbf{u} \cdot \mathbf{T}) + \rho \mathbf{u} \cdot \mathbf{f}] dV \end{aligned}$$

Conservation Equations

Total energy equation

$$\nabla \cdot (\mathbf{u} \cdot \mathbf{T}) = \mathbf{u} \cdot (\nabla \cdot \mathbf{T}) + \mathbf{T} : \nabla \mathbf{u}$$

$$\frac{\partial}{\partial t} \left[\rho \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \right] + \nabla \cdot \left[\rho \mathbf{u} \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \right] = -\nabla \cdot \mathbf{q} + \mathbf{u} \cdot (\nabla \cdot \mathbf{T}) + \rho \mathbf{u} \cdot \mathbf{f}$$

$$\left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \left(\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right) = 0$$

$$\rho \frac{\partial}{\partial t} \left(e + \frac{1}{2} \cancel{\mathbf{u} \cdot \mathbf{u}} \right) + \rho \mathbf{u} \cdot \nabla \left(e + \frac{1}{2} \cancel{\mathbf{u} \cdot \mathbf{u}} \right) = -\nabla \cdot \mathbf{q} + \mathbf{u} \cdot (\nabla \cdot \mathbf{T}) + \cancel{\rho \mathbf{u} \cdot \mathbf{f}}$$

$$\mathbf{u} \cdot \left(\cancel{\rho \frac{\partial \mathbf{u}}{\partial t}} + \cancel{\rho \mathbf{u} \cdot \nabla \mathbf{u}} = \nabla \cdot \mathbf{T} + \cancel{\rho \mathbf{f}} \right)$$

$$\rho \frac{\partial e}{\partial t} + \rho \mathbf{u} \cdot \nabla e = -\nabla \cdot \mathbf{q} + \mathbf{T} : (\nabla \mathbf{u})$$

Conservation Equations

Constitutive Relations

$$\mathbf{q} = -k\nabla T \quad (\text{Fourier's Law})$$

$$p = p(e, \rho), \quad T = T(e, \rho) \quad (\text{Equation of State})$$

If viscous heating is negligible,

$$\mathbf{T} : \nabla \mathbf{u} = -p \mathbf{U} : \nabla \mathbf{u} = -\nabla \cdot (p \mathbf{u}) + \mathbf{u} \cdot \nabla p = -p(\nabla \cdot \mathbf{u})$$

$$\rho \frac{\partial e}{\partial t} + \rho \mathbf{u} \cdot \nabla e = -\nabla \cdot \mathbf{q} - p(\nabla \cdot \mathbf{u})$$

Conservation Equations – Summary

In Convective (Non-conservative) Form

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0$$

$$\rho \frac{D\mathbf{u}}{Dt} = \nabla \cdot \mathbf{T} + \rho \mathbf{f}$$

$$\rho \frac{De}{Dt} = -\nabla \cdot \mathbf{q} + \mathbf{T} : (\nabla \mathbf{u})$$

Conservation Equations – Summary

In Conservative Form

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{u})$$

$$\frac{\partial}{\partial t} (\rho \mathbf{u}) = \nabla \cdot (\mathbf{T} - \rho \mathbf{u} \mathbf{u}) + \rho \mathbf{f}$$

$$\frac{\partial}{\partial t} \left[\rho \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \right] = -\nabla \cdot \left[\rho \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \mathbf{u} - \mathbf{u} \cdot \mathbf{T} + \mathbf{q} \right]$$

Discretized equations can satisfy the conservation properties more easily

Conservation Equations – Summary

In Integral Form (for fixed CV)

$$\frac{\partial}{\partial t} \int_{CV} \rho dV + \int_{CV} \rho \mathbf{u} \cdot \mathbf{n} dS = 0$$

$$\frac{\partial}{\partial t} \int_{CV} \rho \mathbf{u} dV = \int_{CV} \rho \mathbf{f} dV + \int_{CS} [\mathbf{T} \cdot \mathbf{n} - \rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n})] dS$$

$$\begin{aligned} \frac{\partial}{\partial t} \int_{CV} \rho \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) dV &= \int_{CV} \rho \mathbf{u} \cdot \mathbf{f} dV \\ &+ \int_{CS} \left[\mathbf{u} \cdot (\mathbf{n} \cdot \mathbf{T}) - \mathbf{n} \cdot \left(\rho \left(e + \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) + \mathbf{q} \right) \right] dS \end{aligned}$$

Useful in Finite Volume Methods

Conservation Equations – Special Cases

Incompressible Flows

$$\frac{D\rho}{Dt} = 0$$

Continuity equation reduces to

$$\cancel{\frac{1}{\rho} \frac{D\rho}{Dt}} + \nabla \cdot \mathbf{u} = 0 \quad \Rightarrow \quad \nabla \cdot \mathbf{u} = 0 \quad (\text{Divergence-free})$$

Momentum equation with constant viscosity, $\kappa = 0$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

$\nu = \mu/\rho$: kinematic viscosity

Conservation Equations – Special Cases

Equation for Pressure

Take divergence of the momentum equation

$$\nabla \cdot \left[\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f} \right]$$

$$\frac{\partial}{\partial t} (\cancel{\nabla \cdot \mathbf{u}}) + \nabla \cdot (\mathbf{u} \cdot \nabla \mathbf{u}) = -\frac{1}{\rho} \nabla^2 p + \nu \nabla^2 (\cancel{\nabla \cdot \mathbf{u}}) + \nabla \cdot \mathbf{f}$$

$$\nabla^2 p = -\rho \nabla \cdot (\mathbf{u} \cdot \nabla \mathbf{u}) + \rho \nabla \cdot \mathbf{f} \quad : \text{Poisson's equation}$$

which replaces the incompressibility condition ($\nabla \cdot \mathbf{u} = 0$)

We'll come back to this issue later.

Conservation Equations – Special Cases

Compressible inviscid flows

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{E}}{\partial x} + \frac{\partial \mathbf{F}}{\partial y} + \frac{\partial \mathbf{G}}{\partial z} = 0$$

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho E \end{pmatrix} \quad \mathbf{E} = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uw \\ (\rho E + p)u \end{pmatrix} \quad \mathbf{F} = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vw \\ (\rho E + p)v \end{pmatrix} \quad \mathbf{G} = \begin{pmatrix} \rho w \\ \rho uw \\ \rho vw \\ \rho w^2 + p \\ (\rho E + p)w \end{pmatrix}$$

Constitutive relations

$$p = \rho RT, \quad e = c_v T, \quad h = c_p T \quad \text{or} \\ c_v = \frac{R}{\gamma - 1}, \quad p = (\gamma - 1)\rho e, \quad T = \frac{(\gamma - 1)e}{R}$$

Solution Methods for Incompressible Navier-Stokes Equations

Incompressible N-S equations

Governing equations for incompressible flows (no gravity):

$$\nabla \cdot \mathbf{u} = 0$$

$$\frac{\partial \mathbf{u}}{\partial t} = -\underbrace{\mathbf{u} \cdot \nabla \mathbf{u}}_{\text{Advection}} + \underbrace{\nu \nabla^2 \mathbf{u}}_{\text{Diffusion}} - \frac{1}{\rho} \underbrace{\nabla p}_{\text{Pressure}}$$

- The (hydrodynamic) pressure is decomposed from the rest of the solution variables. Physically, it is the pressure that drives the flow, but in practice pressure is solved such that the incompressibility condition is satisfied.
- For incompressible flows, the (hydrodynamic) pressure **projects** the velocity field into an incompressible space.
- Mathematically, the system can be interpreted as differential equations with algebraic constraint.

$$(u, v) \quad \nabla \cdot \mathbf{u} = 0$$

Solution Methods of Incompressible N-S Eqs.

Projection Methods

- MAC method (Harlow and Welch, 1965)
- Projection method (Chorin (1968) and Temam (1969))
- Fractional step method (Kim and Moin, 1985)

Pressure-Correction Methods

- SIMPLE (Patankar & Spalding, 1972)
- SIMPLER (Patankar, 1981)

PISO (Pressure Implicit with Splitting of Operators)

PIMPLE (PISO + SIMPLE)

Projection/ Fractional Step Methods

Marker-and-Cell (MAC) Method

Marker-and-Cell (MAC) Method – Harlow and Welch (1965)

- Originally derived for free surface flows with staggered grid

$$\frac{\partial \mathbf{u}}{\partial t} = -\mathbf{u} \cdot \nabla \mathbf{u} + \nu \nabla^2 \mathbf{u} - \frac{1}{\rho} \nabla p \quad \text{and} \quad \nabla \cdot \mathbf{u} = 0$$

- Explicit integration

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} = -\mathbf{u}^n \cdot \nabla_h \mathbf{u}^n + \nu \nabla_h^2 \mathbf{u}^n - \frac{\nabla_h p}{\rho}$$

- Taking divergence of the momentum equation,

$$\frac{\cancel{\nabla_h \cdot \mathbf{u}^{n+1}} - \cancel{\nabla_h \cdot \mathbf{u}^n}}{\Delta t} + \nabla_h \cdot (\mathbf{u}^n \cdot \nabla_h \mathbf{u}^n) = -\frac{\nabla_h^2 p}{\rho} + \nu \nabla_h^2 (\cancel{\nabla_h \cdot \mathbf{u}^n})$$

$$\nabla_h^2 p = -\rho \nabla_h \cdot (\mathbf{u}^n \cdot \nabla_h \mathbf{u}^n) : \text{Poisson's equation}$$

Projection Method

Projection Method – Chorin (1968), Temam (1969)

- Originally derived on a collocated grid
- Identical to MAC except for the Poisson's equation
- Split

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} = -\mathbf{u}^n \cdot \nabla_h \mathbf{u}^n + \nu \nabla_h^2 \mathbf{u}^n - \frac{\nabla_h p}{\rho}$$

into

$$\frac{\mathbf{u}^t - \mathbf{u}^n}{\Delta t} = -\mathbf{u}^n \cdot \nabla_h \mathbf{u}^n + \nu \nabla_h^2 \mathbf{u}^n \Rightarrow \mathbf{u}^t = \mathbf{u}^n + \Delta t(-\mathbf{A}^n + \mathbf{D}^n)$$

and

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^t}{\Delta t} = -\frac{1}{\rho} \nabla_h p \Rightarrow \mathbf{u}^{n+1} = \mathbf{u}^t - \frac{\Delta t}{\rho} \nabla_h p$$

by introducing the temporary velocity $\mathbf{u}^t$

Projection Method

- To derive an equation for the pressure, we take the divergence

$$\nabla_h \cdot \left(\frac{\mathbf{u}^{n+1} - \mathbf{u}^t}{\Delta t} \right) = -\frac{1}{\rho} \nabla_h \cdot \nabla_h p$$

and use the mass conservation equation

$$\cancel{\nabla_h \cdot \mathbf{u}^{n+1}} = \nabla_h \cdot \mathbf{u}^t - \frac{\Delta t}{\rho} \nabla_h \cdot \nabla_h p$$

- The result is

$$\nabla_h^2 p = \frac{\rho}{\Delta t} \nabla_h \cdot \mathbf{u}^t$$

Splitting integration procedure:

1. Integrate without pressure: $\mathbf{u}^n \rightarrow \mathbf{u}^t$
2. **Project** into a space to enforce incompressibility

$$\mathbf{u}^t \rightarrow \mathbf{u}^{n+1} \quad \text{with} \quad \nabla_h \cdot \mathbf{u}^{n+1} = 0$$

MAC vs. Projection Methods

MAC

vs.

Projection

1. Integrate without pressure

$$\mathbf{u}^n = \mathbf{u}^n$$

$$\mathbf{u}^t = \mathbf{u}^n + \Delta t(-\mathbf{A}^n + \mathbf{D}^n)$$

2. Pressure equation

$$\nabla_h^2 p = -\rho \nabla_h \cdot (\mathbf{u}^n \cdot \nabla_h \mathbf{u}^n)$$

$$\nabla_h^2 p = \frac{\rho}{\Delta t} \nabla_h \cdot \mathbf{u}^t$$

3. Projection into incompressible field

$$\mathbf{u}^{n+1} = \mathbf{u}^n + \Delta t(-\mathbf{A}^n + \mathbf{D}^n) - \frac{\Delta t}{\rho} \nabla_h p$$

$$\mathbf{u}^{n+1} = \mathbf{u}^t - \frac{\Delta t}{\rho} \nabla_h p$$

Predictor – Corrector

Predictor – Corrector

A second-order method is obtained by taking two successive forward Euler steps, then averaging the result with the initial state:

$$\begin{aligned}\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} &= -\nabla \mathbf{u}^n \mathbf{u}^n + \nu \nabla^2 \mathbf{u}^n \\ \nabla^2 P^* &= \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^* \\ \mathbf{u}^1 &= \mathbf{u}^* - \Delta t \nabla P^*\end{aligned}$$

Second forward step using the predicted velocity

$$\begin{aligned}\frac{\mathbf{u}^{**} - \mathbf{u}^1}{\Delta t} &= -\nabla \mathbf{u}^1 \mathbf{u}^1 + \nu \nabla^2 \mathbf{u}^1 \\ \nabla^2 P^{**} &= \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^{**} \\ \mathbf{u}^2 &= \mathbf{u}^{**} - \Delta t \nabla P^{**}\end{aligned}$$

Then average the results: $\mathbf{u}^{n+1} = \frac{1}{2} (\mathbf{u}^1 + \mathbf{u}^2)$ This is Heun's method (explicit trapezoidal rule, RK2).

Adam-Bashford/Crank-Nicolson

Adam-Bashford/Crank-Nicolson

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} = -\frac{1}{2} [3\mathbf{A}(\mathbf{u}^n) - \mathbf{A}(\mathbf{u}^{n-1})] + \frac{\nu}{2} \nabla^2 (\mathbf{u}^n + \mathbf{u}^{n+1}) - \nabla P$$

$$\nabla \cdot \mathbf{u}^{n+1} = 0 \quad \text{Adams-Bashforth (AB2)} \quad \text{Crank-Nicolson}$$

Split:

$$\frac{\tilde{\mathbf{u}} - \mathbf{u}^n}{\Delta t} = -\frac{1}{2} [3\mathbf{A}(\mathbf{u}^n) - \mathbf{A}(\mathbf{u}^{n-1})] + \frac{\nu}{2} \nabla^2 \mathbf{u}^n$$

$$\left. \begin{aligned} \frac{\mathbf{u}^{n+1} - \tilde{\mathbf{u}}}{\Delta t} &= -\nabla P + \frac{\nu}{2} \nabla^2 \mathbf{u}^{n+1} \\ \nabla \cdot \mathbf{u}^{n+1} &= 0 \end{aligned} \right\} \nabla^2 P = \frac{1}{\Delta t} \nabla \cdot \tilde{\mathbf{u}}$$

The correction equation is implicit and must be solved by an iteration in the same way as the pressure equation

Fractional Step Method with RK Time Integration

A complete Runge-Kutta time integration (Weinan E.)

First a half step:
$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t/2} = -\nabla \mathbf{u}^n \mathbf{u}^n + \nu \nabla^2 \mathbf{u}^n$$

$$\nabla^2 P^* = \frac{2}{\Delta t} \nabla \cdot \mathbf{u}^*$$

$$\mathbf{u}^1 = \mathbf{u}^* - \frac{\Delta t}{2} \nabla P^*$$

Continue for the second step

$$\frac{\mathbf{u}^{**} - \mathbf{u}^n}{\Delta t/2} = -\nabla \mathbf{u}^1 \mathbf{u}^1 + \nu \nabla^2 \mathbf{u}^1$$

$$\nabla^2 P^{**} = \frac{2}{\Delta t} \nabla \cdot \mathbf{u}^{**}$$

$$\mathbf{u}^2 = \mathbf{u}^{**} - \frac{\Delta t}{2} \nabla P^{**}$$

Fractional Step Method with RK Time Integration

A complete Runge-Kutta time integration (continued)

Take a full step
using the predicted
velocity

$$\frac{\mathbf{u}^{***} - \mathbf{u}^n}{\Delta t} = -\nabla \mathbf{u}^2 \mathbf{u}^2 + \nu \nabla^2 \mathbf{u}^2$$

$$\nabla^2 P^{***} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^{***}$$

$$\mathbf{u}^3 = \mathbf{u}^{***} - \Delta t \nabla P^{***}$$

Then compute

$$k = \Delta t (-\nabla \mathbf{u}^3 \mathbf{u}^3 + \nu \nabla^2 \mathbf{u}^3)$$

And finally

$$\mathbf{u}^+ = \frac{1}{3} (-\mathbf{u}^n + \mathbf{u}^1 + 2\mathbf{u}^2 + \mathbf{u}^3) + \frac{1}{6} k$$

$$\nabla^2 P^+ = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^+$$

$$\mathbf{u}^{n+1} = \mathbf{u}^+ - \Delta t \nabla P^+$$

Pressure Correction Methods

Pressure Correction Method

SIMPLE Algorithm – Patankar & Spalding (1972)

(**S**emi-**I**mplicit **M**ethod for **P**ressure **L**inked **E**quations)

- Iterative procedure with pressure correction $p = p_0 + p'$

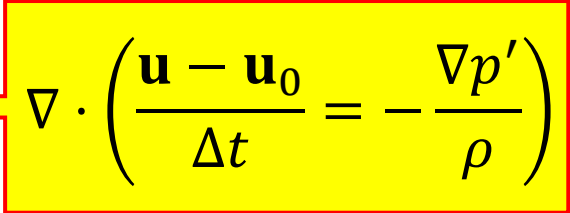
1. Guess the pressure field p_0
2. Solve the momentum equation (implicitly)

$$\frac{\mathbf{u}_0 - \mathbf{u}^n}{\Delta t} = -\mathbf{u}_0 \cdot \nabla \mathbf{u}_0 + \nu \nabla^2 \mathbf{u}_0 - \frac{\nabla p_0}{\rho}$$

3. Solve the pressure correction equation

$$\nabla^2 p' = \frac{\rho}{\Delta t} (\nabla \cdot \mathbf{u}_0)$$

Exact only if $\mathbf{u}' = 0$


$$\nabla \cdot \left(\frac{\mathbf{u} - \mathbf{u}_0}{\Delta t} = -\frac{\nabla p'}{\rho} \right)$$

Pressure Correction Method

4. Correct the pressure and velocity

$$p = p_0 + p'$$

$$\mathbf{u} = \mathbf{u}_0 - \frac{\Delta t}{\rho} \nabla p'$$

5. Go to 2. Repeat the process until the solution converges

Notes:

- Originally developed for the staggered grid system
- The corrected velocity field satisfies the continuity equation even if the pressure correction is only approximate
- Sometimes p' tends to be overestimated

$$p = p_0 + \omega p' \quad (\omega \approx 0.8) \text{ underrelaxation}$$

Pressure Correction Method

Derivation of Poisson's Equation

$$p = p_0 + p', \mathbf{u} = \mathbf{u}_0 + \mathbf{u}'$$

$$\frac{\mathbf{u}_0 - \mathbf{u}^n}{\Delta t} = -\mathbf{u}_0 \cdot \nabla \mathbf{u}_0 + \nu \nabla^2 \mathbf{u}_0 - \frac{\nabla p_0}{\rho} : \text{Guessed equation}$$

$$\frac{\mathbf{u} - \mathbf{u}^n}{\Delta t} = -\mathbf{u} \cdot \nabla \mathbf{u} + \nu \nabla^2 \mathbf{u} - \frac{\nabla p}{\rho} : \text{Converged equation}$$

Subtracting and assuming $\mathbf{u} \cdot \nabla \mathbf{u} = \mathbf{u}_0 \cdot \nabla \mathbf{u}_0$, $\nu \nabla^2 \mathbf{u} = \nu \nabla^2 \mathbf{u}_0$

$$\nabla \cdot \left(\frac{\mathbf{u} - \mathbf{u}_0}{\Delta t} = -\frac{\nabla p'}{\rho} \right) \Rightarrow \nabla^2 p' = \frac{\rho}{\Delta t} (\nabla \cdot \mathbf{u}_0)$$

with $(\nabla \cdot \mathbf{u} = 0)$

Pressure Correction Method

SIMPLER (SIMPLE Revised) – Patankar (1981)

- Incorporating the projection method (fractional step)

1. Guess the velocity field $\mathbf{u}_0$
2. Solve the momentum equation **without pressure**

$$\frac{\hat{\mathbf{u}} - \mathbf{u}^n}{\Delta t} = -\mathbf{u}_0 \cdot \nabla \mathbf{u}_0 + \nu \nabla^2 \mathbf{u}_0$$

3. Solve the pressure Poisson equation

$$\nabla^2 p^* = \frac{\rho}{\Delta t} (\nabla \cdot \hat{\mathbf{u}})$$

Pressure Correction Method

4. Solve the momentum equation with p^* implicitly

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = -\mathbf{u}^* \cdot \nabla \mathbf{u}^* + \nu \nabla^2 \mathbf{u}^* - \frac{1}{\rho} \nabla p^*$$

5. Pressure correction equation

$$\nabla^2 p' = \frac{\rho}{\Delta t} (\nabla \cdot \mathbf{u}^*)$$

6. Correct the velocity, but not the pressure

$$\mathbf{u} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p'$$

7. Go to 2. Repeat the process until solution is converged

Pressure Correction Method

Note on SIMPLER Algorithm

- If the initial velocity field $\mathbf{u}_0$ is a correct one, then p^* is also the correct one, so no further iteration is needed
- Then the difference between SIMPLER and projection method is simply the time integration scheme

SIMPLER – implicit, iterative

commonly used in steady problems

Projection – explicit or factorization (Kim & Moin)

well suited for unsteady problems

- Iterative procedure can benefit from the multi-grid method

PISO (pressure-implicit with splitting of operators) – Issa (1986)

1. Predictor step – using p^n (same as SIMPLE)

$$\frac{\rho}{\Delta t} (\mathbf{u}^* - \mathbf{u}^n) = H(\mathbf{u}^*) - \nabla p^n + S_i \quad (\text{implicit})$$

$$\nabla \cdot \mathbf{u}^* \neq 0 \quad \text{in general}$$

2. First corrector step

(derive) 

$$\nabla^2 p^* = \nabla \cdot H(\mathbf{u}^*) + \nabla \cdot S_i + \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^n$$
$$\left\{ \begin{array}{l} \frac{\rho}{\Delta t} (\mathbf{u}^{**} - \mathbf{u}^n) = H(\mathbf{u}^*) - \nabla p^* + S_i \quad (\text{explicit}) \\ \nabla \cdot \mathbf{u}^{**} = 0 \end{array} \right.$$

3. Second corrector step

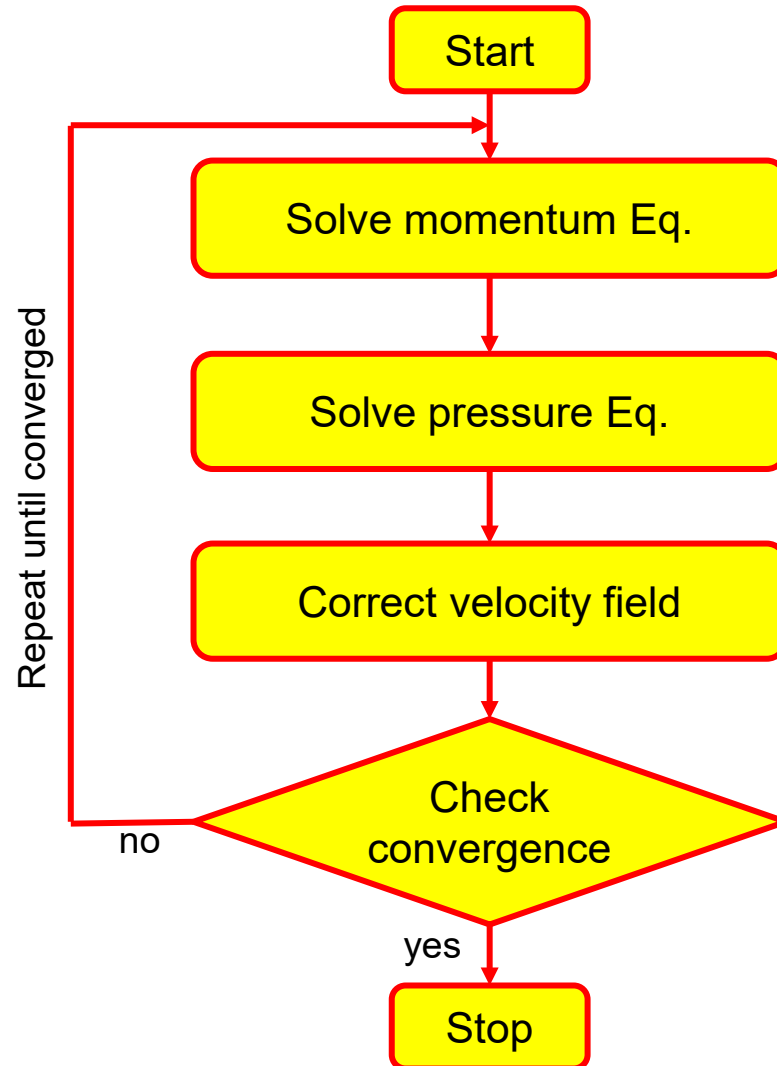
(derive) 

$$\nabla^2 p^{**} = \nabla \cdot H(\mathbf{u}^{**}) + \nabla \cdot S_i + \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^n$$
$$\left\{ \begin{array}{l} \frac{\rho}{\Delta t} (\mathbf{u}^{***} - \mathbf{u}^n) = H(\mathbf{u}^{**}) - \nabla p^{**} + S_i \quad (\text{explicit}) \\ \nabla \cdot \mathbf{u}^{***} = 0 \end{array} \right.$$

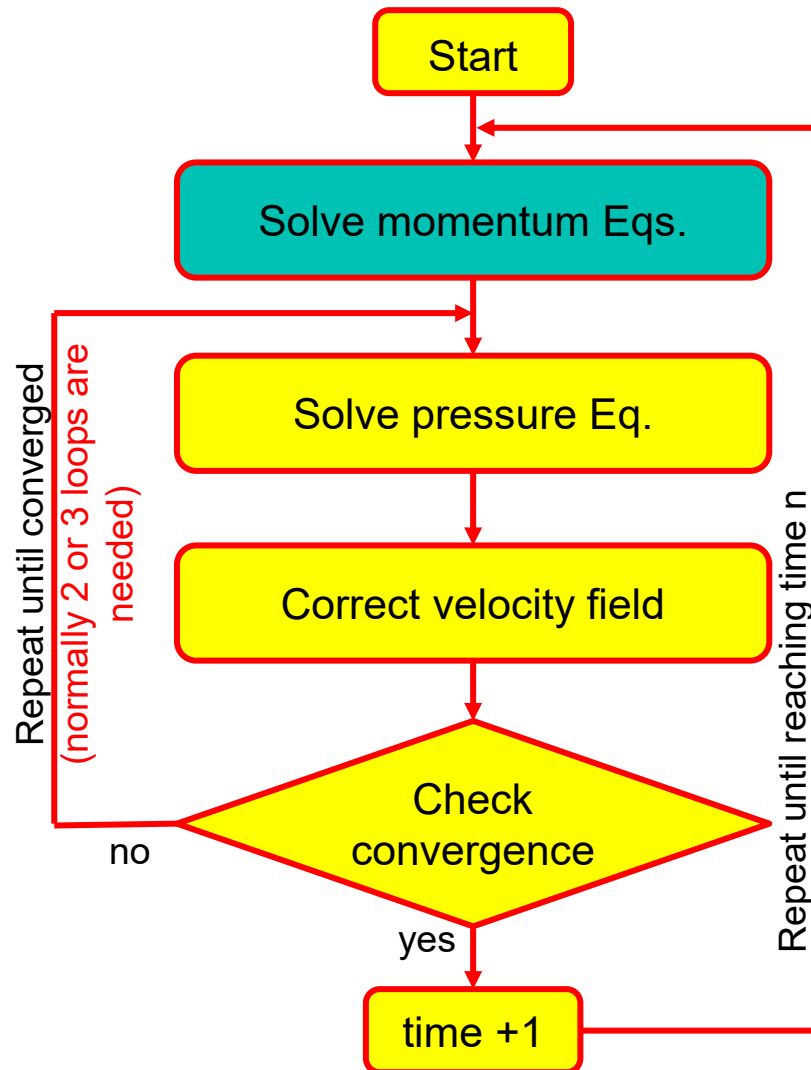
More correction can be made, but not necessary because splitting errors for the first and second steps are sufficiently small

Algorithms for SIMPLE, PISO, and PIMPLE

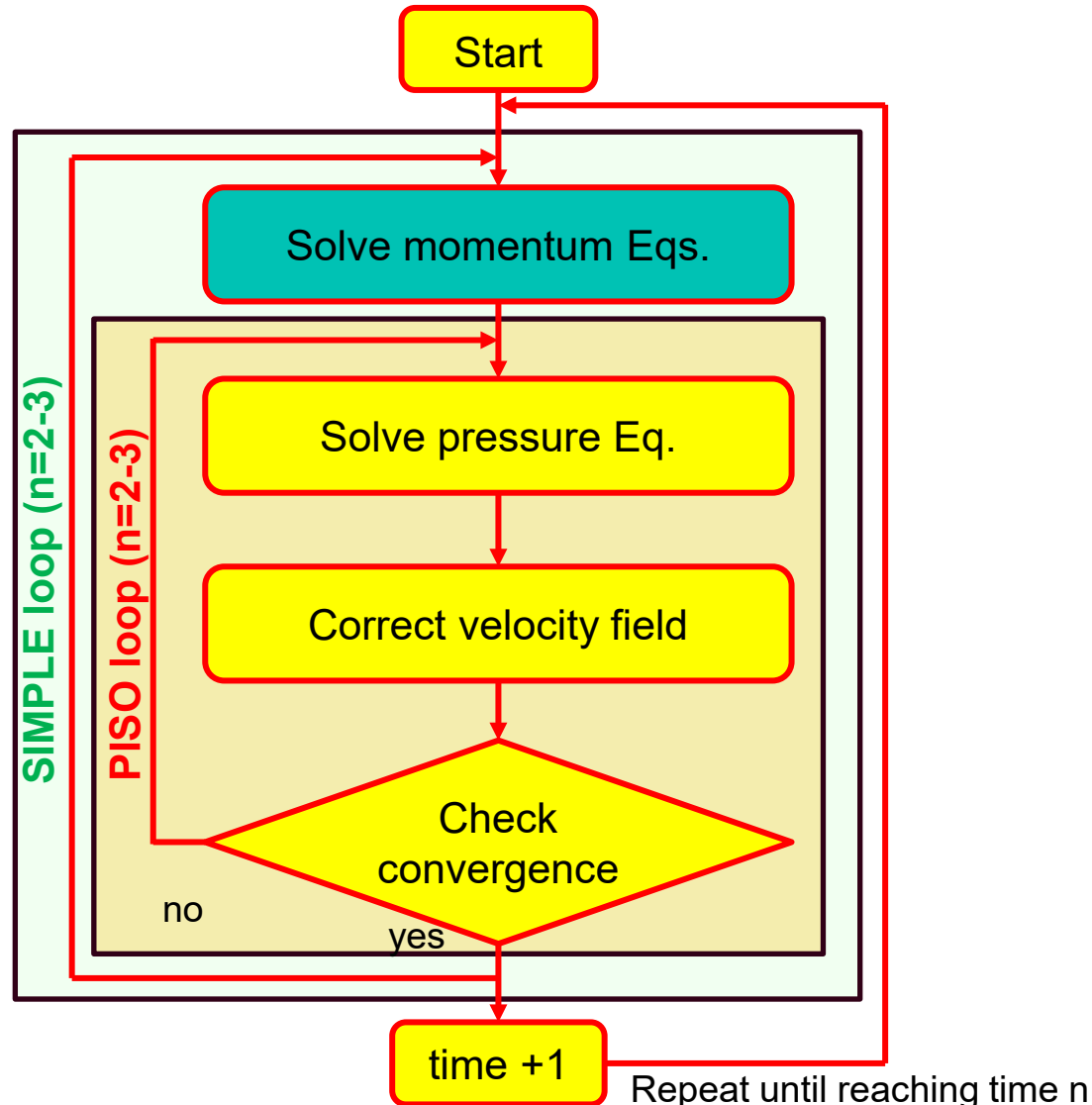
SIMPLE Algorithm



PISO Algorithm



PIMPLE Algorithm



Comparison

- **SIMPLE vs. PISO vs. PIMPLE:**

Features	SIMPLE	PISO	PIMPLE
Best for	Steady	Transient	Steady/transient
Iteration type	Outer only	No outer loop	Outer + inner
Pressure correction	One per step	Multiple	Multiple
Under-relaxation	Required	Not needed	Sometimes needed

Compressible Navier-Stokes Equations

Compressible N-S Equations

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{E}}{\partial x} + \frac{\partial \mathbf{F}}{\partial y} + \frac{\partial \mathbf{G}}{\partial z} = 0$$

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho E \end{pmatrix} \quad \mathbf{E} = \begin{pmatrix} \rho u \\ \rho u^2 + p - \tau_{xx} \\ \rho uv - \tau_{xy} \\ \rho uw - \tau_{xz} \\ (\rho E + p)u + \psi_x \end{pmatrix} \quad \mathbf{F} = \begin{pmatrix} \rho v \\ \rho uv - \tau_{xy} \\ \rho v^2 + p - \tau_{yy} \\ \rho vw - \tau_{yz} \\ (\rho E + p)v + \psi_y \end{pmatrix} \quad \mathbf{G} = \begin{pmatrix} \rho w \\ \rho uw - \tau_{xz} \\ \rho vw - \tau_{yz} \\ \rho w^2 + p - \tau_{zz} \\ (\rho E + p)w + \psi_z \end{pmatrix}$$

where

Constitutive relations

$$\psi_x = -u\tau_{xx} - v\tau_{xy} - w\tau_{xz} + q_x$$

$$\psi_y = -u\tau_{xy} - v\tau_{yy} - w\tau_{yz} + q_y$$

$$\psi_z = -u\tau_{xz} - v\tau_{yz} - w\tau_{zz} + q_z$$

$$p = \rho RT, \quad e = c_v T, \quad h = c_p T \quad \text{or}$$

$$c_v = \frac{R}{\gamma - 1}, \quad p = (\gamma - 1)\rho e, \quad T = \frac{(\gamma - 1)e}{R}$$

Compressible N-S Equations

Solution methods for compressible N-S equations

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{E}}{\partial x} + \frac{\partial \mathbf{F}}{\partial y} + \frac{\partial \mathbf{G}}{\partial z} = 0$$

follows the same techniques used for hyperbolic equations

For smooth solutions with viscous terms, central differencing usually works

- No need to worry about upwind method, flux-splitting, TVD, FCT (flux-corrected transport), etc.
- In general, upwind-like methods introduce numerical dissipation, hence provide stability, but accuracy becomes a concern.

Compressible N-S Equations

Explicit methods

- MacCormack method
- Leapfrog/DuFort-Frankel method
- Lax-Wendroff method
- Runge-Kutta method

Implicit methods

- Beam-Warming scheme
- Runge-Kutta method

More methods are 2nd order

The Runge-Kutta method can be easily tailored to higher order method (both implicit and explicit)

Compressible N-S Equations

Most of the time, an implicit integration method involves nonlinear advection terms

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{E}}{\partial x} + \frac{\partial \mathbf{F}}{\partial y} + \frac{\partial \mathbf{G}}{\partial z} = 0$$

which are linearized as

$$\frac{\mathbf{U}^{n+1} - \mathbf{U}^n}{\Delta t} = - \left([A] \frac{\partial \mathbf{U}^{n+1}}{\partial x} + [B] \frac{\partial \mathbf{U}^{n+1}}{\partial y} + [C] \frac{\partial \mathbf{U}^{n+1}}{\partial z} \right)$$

$$[A] = \left(\frac{\partial \mathbf{E}}{\partial \mathbf{U}} \right)^n, [B] = \left(\frac{\partial \mathbf{F}}{\partial \mathbf{U}} \right)^n, [C] = \left(\frac{\partial \mathbf{G}}{\partial \mathbf{U}} \right)^n$$

+ ADI, factorization, etc.

Compressible N-S Equations

Ultimately, compressible Navier-Stokes equations can be written as a system of ODE's

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{E}}{\partial x} + \frac{\partial \mathbf{F}}{\partial y} + \frac{\partial \mathbf{G}}{\partial z} = 0$$

$$\Rightarrow \frac{d\mathbf{U}}{dt} = F(t, \mathbf{U}(t))$$

$$\mathbf{U}(t_0) = \mathbf{U}_0 \quad \text{Initial conditions}$$

Solution techniques for a system of ODE applies

- Explicit vs. Implicit (non-stiff vs. stiff)
- One-step vs. Multi-step

Boundary Conditions for Compressible Flows

Boundary conditions for the compressible flows are trickier

- All the acoustic waves must be properly taken care of at the boundaries

Typical boundary conditions:

- Periodic: still easy to implement
- Both inflow and outflow conditions require treatment of characteristic waves (hard-wall, nonreflecting, sponge, etc)
- NSCBC (Navier-Stokes Characteristic Boundary Conditions)
 - LODI (local one-dimensional inviscid) relations (Poinsot & Lele, 1992)
 - 3D-NSCBC: Improve the original LODI relations (Yoo et al. (2005), Yoo & Im (2007), Lodato et al. (2008))